

Temporal Analysis of Host Transcriptional Responses to *Chlamydia muridarum* Infection Following CPAF-Based Vaccination in a Mouse Model

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Abstract

Chlamydia trachomatis is the most prevalent bacterial sexually transmitted infection globally, yet no licensed vaccine currently exists, and the local immune mechanisms underlying vaccine-induced protection remain poorly understood. In this study, probe-based dual RNA sequencing of longitudinal cervical swab samples was used to simultaneously characterize host and pathogen gene expression in CPAF-vaccinated and naive mice at -2 , 4, 6, and 11 days post-infection (dpi).

Vaccinated mice exhibited lower chlamydial gene counts at all post-infection time points, indicating reduced bacterial burden. At baseline (-2 dpi), no significant differences in host gene expression were observed between groups. Following infection, gene expression patterns diverged: vaccinated mice mounted an earlier and more focused immune response that peaked at 6 dpi and declined by 11 dpi, consistent with a vaccine-induced recall response. In contrast, naive mice showed a less pronounced response at 4 dpi but stronger inflammatory gene expression at later time points, suggesting delayed control of infection and prolonged inflammatory signaling. Several differentially expressed genes, including *S100a8*, *S100a9*, *Il17a*, and *Cd3e*, were identified as candidate transcriptional markers associated with coordinated innate and adaptive immune activation in vaccinated mice.

These findings establish a longitudinal swab-based transcriptional framework for evaluating mucosal vaccine responses and identify candidate host and pathogen markers that may inform the

design of future chlamydia vaccine studies.

Introduction

Chlamydia trachomatis (CT) is the most prevalent bacterial sexually transmitted infection worldwide, with over 128 million new cases estimated annually [WHO, 2024]. A substantial proportion of infections are asymptomatic, approximately 70% in women and 50% in men, which contributes to delayed diagnosis, continued transmission, and frequent reinfection [Mabonga et al., 2021]. Untreated infection can progress to pelvic inflammatory disease (PID), causing tubal scarring, infertility, and increased risk of ectopic pregnancy. In the United States alone, over 1.5 million cases were reported in 2024 [CDC, 2024]. Chlamydia and gonorrhea together impose an estimated \$1 billion annually in direct medical costs in the United States, driven largely by these complications [Kumar et al., 2021].

Despite CDC recommendations for routine screening, infection rates remain high [CDC, 2025], indicating that screening alone is insufficient for sustained disease control. Vaccination represents a more durable long-term prevention strategy. Early vaccine efforts using whole inactivated organisms provided only short-term, strain-specific protection. In some cases, they also induced adverse immune responses including delayed hypersensitivity and increased susceptibility to reinfection, suggesting the elicited immunity was incomplete [Karunakaran et al., 2010]. These limitations drove the development of subunit vaccines targeting specific chlamydial antigens designed to induce more precise and protective responses.

Among these, the major outer membrane protein (MOMP) has been widely studied due to its strong immunogenicity and ability to induce both neutralizing antibody and T cell responses [Kari et al., 2009]. However, high antigenic variability across CT serovars limits its potential for broad, durable protection [Kari et al., 2009, Broder et al., 2024]. In contrast, the chlamydial protease-like activity factor (CPAF) is highly conserved across serovars and abundantly expressed during infection, making it an attractive candidate for inducing cross-protective immune responses [Murthy et al., 2009, Schautteet et al., 2011]. CPAF-based vaccination has shown promise in animal models, particularly when combined with adjuvants that promote strong cellular immunity. More recent studies have demonstrated that CPAF-based vaccines are highly immunogenic in translational animal models such as pigs [Proctor et al., 2024]. STING agonists such as ADU-S100 enhance in-

nate immune signaling and promote robust systemic and mucosal immune responses, including T cell activation [Gogoi et al., 2020].

Despite these advances, the local mucosal immune responses induced by CPAF-based vaccination remain incompletely characterized. Prior studies have demonstrated that protective immunity against *Chlamydia* relies on CD4⁺ T cell-mediated responses and tissue-resident memory formation at mucosal sites [Stary, 2015]. However, these studies primarily rely on systemic immune readouts or endpoint tissue analyses [Morrison and Caldwell, 2002], which do not capture genome-wide transcriptional responses at the site of infection across key stages of disease progression. As a result, it remains unclear how CPAF + ADU-S100 vaccination shapes local transcriptional responses in the genital tract, or whether these responses differ from those in unvaccinated hosts in ways that meaningfully correlate with protection.

Cervical swab sampling provides a minimally invasive approach to capture host and pathogen signals directly from the site of infection [Meyer, 2016]. When combined with probe-based dual RNA sequencing, which enables simultaneous quantification of host and bacterial transcripts from low-biomass samples, this approach allows longitudinal characterization of host-pathogen interactions without terminal sampling.

In this study, we used a mouse model of *Chlamydia muridarum* infection to compare transcriptional responses between CPAF + ADU-S100 vaccinated and naive mice. By analyzing cervical swab samples collected at multiple time points, we aimed to characterize stage-specific local host and pathogen gene expression and identify candidate transcriptional signatures associated with vaccine-induced protection. This study also demonstrates the feasibility of using longitudinal swab-based dual RNA sequencing to study mucosal immune responses during infection.

Methods

Study Design and Experimental Workflow

The experimental design and infection model have been previously described [Poston et al., 2025]. Briefly, female C57BL/6J mice (8–12 weeks old) were assigned to vaccinated and naive groups (n = 10 per group). Vaccinated mice received a prime-boost intranasal mucosal immunization consisting of recombinant *Chlamydia muridarum* CPAF antigen adjuvanted with CpG1826 and the STING

agonist ADU-S100 (Figure 1).

Thirty days after the booster immunization, mice were challenged intravaginally with *Chlamydia muridarum* strain CM006. Cervical swab samples were collected longitudinally at four time points: -2, 4, 6, and 11 days post-infection (dpi), and were used for downstream RNA sequencing to characterize host and pathogen transcriptional responses.

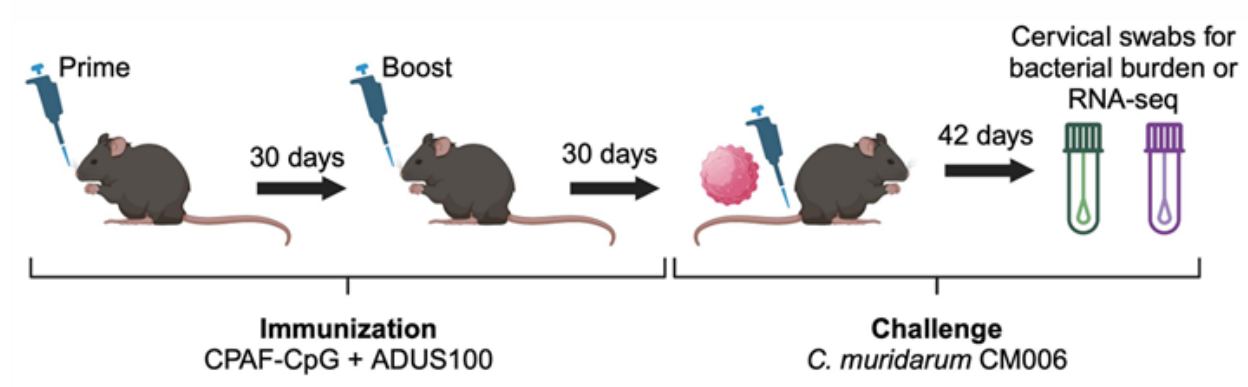


Figure 1: Study design of the *Chlamydia muridarum* vaccination and challenge model. Mice received a prime–boost intranasal immunization with CPAF-CpG + ADU-S100, followed by intravaginal challenge with CM006. Cervical swabs were collected at -2, 4, 6, and 11 dpi for bacterial burden assessment and RNA sequencing.

RNA Extraction, Library Preparation, and Sequencing

RNA was extracted from cervical swab samples preserved in DNA/RNA Shield (Zymo Research) using the KingFisher RNA extraction system (Thermo Fisher Scientific). Given the low-input nature of swab-derived RNA, a probe-based dual RNA sequencing (dual-seq) approach was employed to enable simultaneous profiling of host and pathogen transcripts.

A custom probe library targeting host mRNA and selected *Chlamydia muridarum* genes was used in combination with template-switching chemistry to generate full-length cDNA. Poly-T priming was used to capture host transcripts, while gene-specific probes enabled targeted detection of bacterial transcripts, including rRNA and key developmental markers.

Unique molecular identifiers (UMIs; 12 bp) were incorporated during cDNA synthesis to enable molecule-level quantification and reduce amplification bias. Sequencing libraries were prepared using the NEBNext library preparation kit (New England Biolabs) in a 96-well format and sequenced on an Illumina NextSeq 2000 platform using paired-end reads (28 × 100 cycles).

Data Preprocessing

Each sequencing library generated paired-end FASTQ files, where Read 1 (R1) contained sequence information used for extraction of UMIs, and Read 2 (R2) contained the corresponding cDNA sequences for alignment. Sequencing was performed across two lanes per sample to increase read depth.

Quality control of raw sequencing data was conducted using FastQC to assess read quality and identify potential artifacts, including poly-A/T tails introduced during library preparation. UMIs were extracted from the first 12 bases of R1 reads, and sample-specific barcode sequences were appended to enable compatibility with downstream analysis tools.

Following preprocessing, the modified R1 files and corresponding R2 files were processed using Cell Ranger. All samples were sequenced across two lanes, and the corresponding L001 and L002 FASTQ files were supplied together for each sample and combined during Cell Ranger processing. Reads were aligned to a combined reference genome consisting of mouse GRCm39 and CM sequences, generated using Cell Ranger `mkref` v8.0.0. Gene-level count matrices were generated for each sample.

Although Cell Ranger is designed primarily for single-cell RNA-seq data, it was used here as a preprocessing and UMI-counting tool for bulk cervical swab libraries, given its efficient UMI deduplication and compatibility with custom reference genomes built via Cell Ranger `mkref`. Because each library represented a bulk sample rather than individually barcoded cells, the resulting count matrix was interpreted at the sample level, and no cell-level clustering or annotation was performed. UMIs were used to reduce amplification bias during quantification, while sample barcode processing supported sample assignment rather than single-cell identification.

Quality Control and Sample Assessment

Quality control metrics were extracted from Cell Ranger web summary outputs for each sample, including total number of reads and percentage of reads mapped confidently to the mouse genome. These metrics were used to assess sequencing depth (total number of reads per sample, expressed in millions) and mapping quality across experimental groups (Naive and Vaccinated) and time points (-2, 4, 6, and 11 dpi), and were visualized in Figure 2.

To further assess sample quality, pairwise sample correlations were calculated using $\log_2$ counts

per million (logCPM)-normalized mouse gene expression values. Raw mouse gene counts were first filtered to remove genes with zero counts across all samples, followed by library size normalization and $\log_2$ transformation using the `cpm` function from the `edgeR` package with a prior count of 1 to stabilize variance and avoid undefined values for low or zero counts [Robinson et al., 2010]. Pearson and Spearman correlation matrices were then generated to assess overall similarity in gene expression profiles across samples. The correlation matrices were visualized as hierarchical clustering heatmaps with complete linkage to evaluate global sample similarity by experimental group and time point, and to identify potential outlier samples (Figure 3 and Supplementary Figure S1). Pearson and Spearman correlations were used to provide complementary assessments of sample similarity.

Sample M52_D4 was excluded during the quality-control stage, prior to any downstream analyses, based on two complementary indicators: a low percent of reads mapped confidently to the mouse genome (69.3%; Figure 2) and consistently lower pairwise correlations with other samples. In the logCPM-based Pearson correlation heatmap, M52_D4 showed reduced similarity to the remaining samples, with correlations approaching 0.75–0.80 compared with the higher correlations observed among most other samples (Figure 3). Together, these metrics suggested potential technical failure during library preparation for this sample. To avoid introducing technical noise, M52_D4 was excluded from all subsequent analyses.

Exploratory Data Analysis

Following quality control and sample filtering, chlamydial RNA counts were extracted from the gene-level count matrices generated from the dual RNA-seq libraries. For each targeted CM gene, total read counts and the number of samples with detectable expression (read count > 0) were calculated. These summaries were computed across all samples and stratified by experimental group and time point. Gene-level chlamydial read counts and detection frequencies are presented in Table 1, with stratified summaries shown in Table 2. Highly expressed mouse genes across time points were visualized using a jittered scatter plot (Supplementary Figure S2) based on average raw read counts, with genes exceeding 15,000 counts labeled for visualization and the top 10 genes reported in Supplementary Table S1.

To assess global patterns of gene expression variation across samples, principal component analysis (PCA) was performed on rlog-transformed count data using the DESeq2 package [Love et al., 2014].

Genes were ranked by variance across samples after rlog transformation, and the top 500 most variable genes were used to compute principal components. The first two principal components (PC1 and PC2) were extracted for visualization (Figure 4) and subsequent statistical testing. PCA results were visualized using scatter plots, with samples colored by experimental group and shaped by time point.

To assess the association between experimental group and time point with global gene expression patterns, multivariate analysis of variance (MANOVA) was performed using PC1 and PC2 as joint response variables, and group and day as explanatory variables. Univariate ANOVA models were also applied to PC1 and PC2 separately. Given the exploratory nature of the analysis, MANOVA results were interpreted descriptively based on the reduced-dimensional PC representation rather than as formal confirmatory tests. All analyses were performed in R.

Differential Expression Analysis

Differential gene expression analysis was performed separately at each time point (−2, 4, 6, and 11 dpi) using the DESeq2 package [Love et al., 2014]. This time-point-stratified approach was chosen to characterize stage-specific differences between vaccinated and naive mice. It does not directly model temporal trajectories or group-by-time interactions, which are addressed in the limitations. For each time point, samples were subset accordingly, and genes with zero counts across all samples were removed prior to analysis. Gene expression was modeled as a function of experimental group (naive vs. vaccinated), with the naive group specified as the reference level.

DESeq2 was used to estimate size factors, dispersion parameters, and model coefficients. Log2 fold changes (vaccinated vs. naive) were estimated, and shrinkage was applied using the `apeglm` method [Zhu et al., 2018] to improve stability of effect size estimates, particularly for low-count genes.

Statistical significance was assessed using Wald tests, and p-values were adjusted for multiple testing using the Benjamini–Hochberg method [Benjamini and Hochberg, 1995] to control the false discovery rate (FDR). Genes with adjusted p-values ($p_{adj} < 0.05$) and absolute $\log_2$ fold change greater than 1 were considered significantly differentially expressed.

Volcano plots (Figure 5) were generated for post-infection time points (days 4, 6, and 11) using the `ggplot2` package to visualize differential expression results, with $\log_2$ fold change on the x-axis

and $-\log_{10}(\text{p-value})$ on the y-axis. The top upregulated and downregulated genes at each time point were identified based on ranked $\log_2$ fold change and reported in Supplementary Tables S3–S5.

All analyses were performed in R (version 4.4.0) using DESeq2 (version 1.46.0), with visualization conducted using ggplot2.

Results

The dataset consists of RNA-seq samples collected from vaccinated and naive mice at four time points (−2, 4, 6, and 11 dpi). A total of 20 mice were included (10 per group), with one sample (M52_D4) excluded following quality control, resulting in 79 samples for downstream analysis.

Quality Control

Quality control metrics showed that sequencing depth was generally consistent across samples and within an acceptable range (Figure 2). One sample, M52_D4, was identified during quality control as having lower mapping quality and lower pairwise correlations with other samples, and was therefore excluded from downstream analyses as described in the Methods.

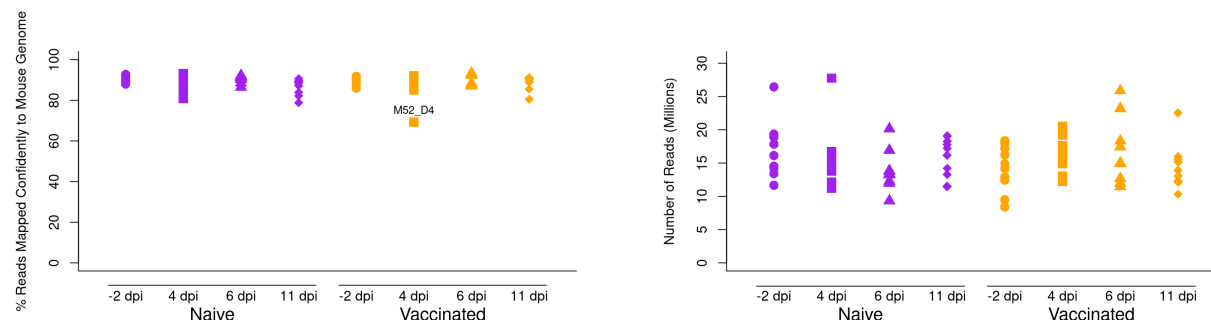


Figure 2: Quality control metrics for RNA-seq data. (A) Percentage of reads confidently mapped to the mouse genome. (B) Sequencing depth across samples.

Both Pearson (Figure 3) and Spearman (Supplementary Figure S1) correlation heatmaps showed high overall similarity among the remaining samples, consistent with expectations given that all samples were derived from the same mouse strain and anatomical site. Hierarchical clustering revealed that samples grouped primarily by time point, with −2 dpi samples forming the most distinct cluster, indicating that pre-infection expression profiles were the most homogeneous across

the dataset.

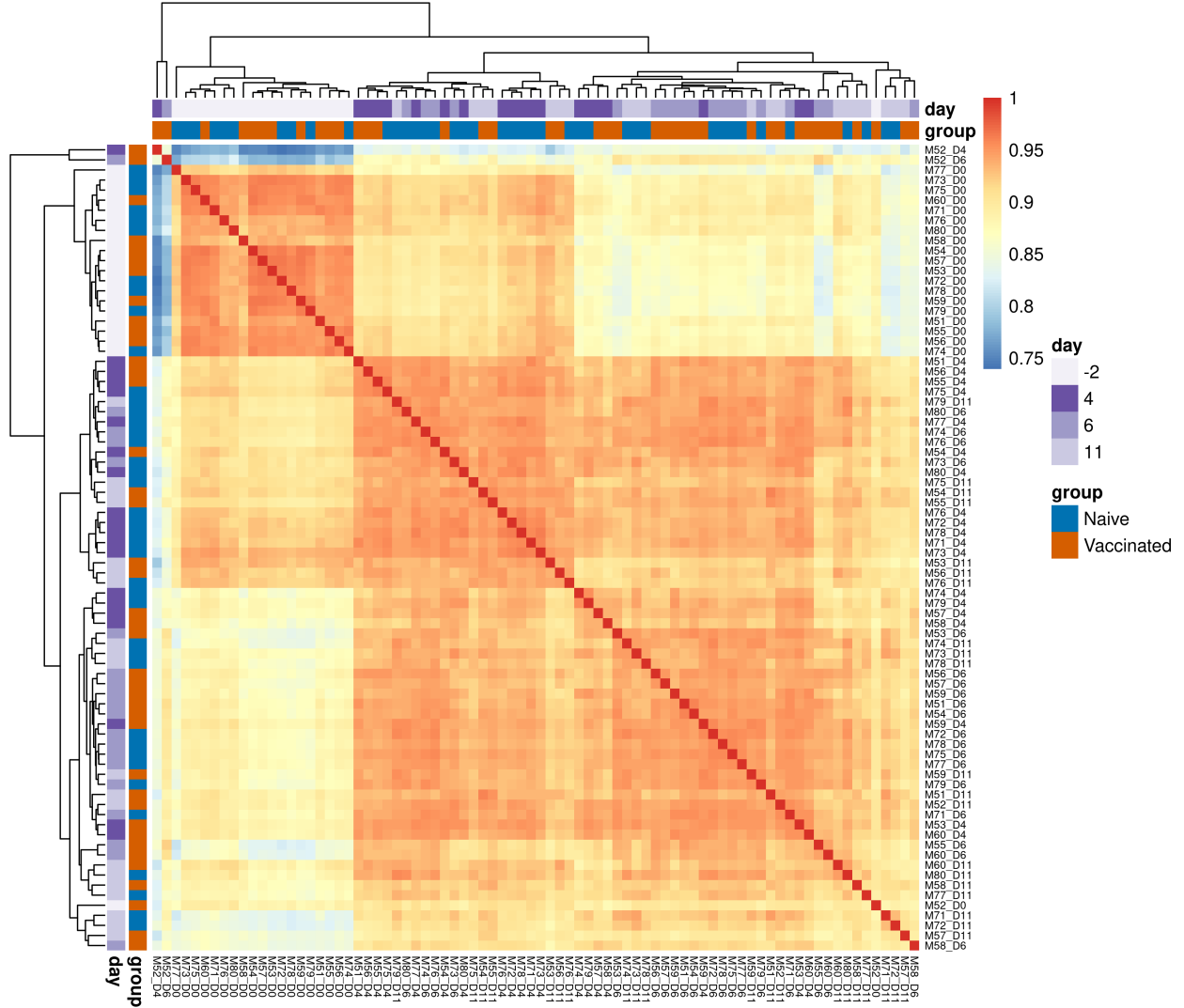


Figure 3: Sample-to-sample Pearson correlation heatmap based on $\log_2$ CPM-normalized gene expression data. Samples cluster primarily by time point, with high overall correlation observed across samples.

Descriptive Analysis

Chlamydial gene-level read counts and detection frequencies across samples are shown in Table 1. The naive group exhibited substantially higher chlamydial gene counts compared to the vaccinated group across all targeted genes, which is consistent with reduced bacterial burden in vaccinated mice. Notably, transcripts from *pmpF* and *CT_858* were rarely detected across samples, suggesting low transcript abundance or limited probe efficiency. As shown in Table 2, chlamydial gene counts

peaked at 4 dpi in both groups, followed by a decrease at 6 dpi and a further reduction by 11 dpi. At -2 dpi, low levels of 23SrRNA and 16SrRNA were detected in both groups despite the absence of infection. These signals may reflect background signal or the presence of transcripts from other bacteria rather than true *Chlamydia*-specific expression.

Table 1: Chlamydial gene-level read counts and detection frequencies across all samples, stratified by group.

Gene	Total		Naive		Vaccinated	
	Read Count	# Libraries	Read Count	# Libraries	Read Count	# Libraries
23SrRNA	10565	65/79	9942	35/40	623	30/39
16SrRNA	8248	69/79	7353	36/40	895	33/39
omcB	693	40/79	647	27/40	46	13/39
hctB	39	16/79	34	13/40	5	3/39
ompA	30	9/79	30	9/40	0	0/39
groEL_1	15	9/79	14	8/40	1	1/39
ftsH	10	7/79	10	7/40	0	0/39
pmpF	6	5/79	6	5/40	0	0/39
CT_858	2	2/79	2	2/40	0	0/39

Table 2: Chlamydial gene-level read counts and detection frequencies stratified by group and time point.

Gene	Naive								Vaccinated							
	-2 dpi		4 dpi		6 dpi		11 dpi		-2 dpi		4 dpi		6 dpi		11 dpi	
	Read	# Mice	Read	# Mice	Read	# Mice	Read	# Mice	Read	# Mice	Read	# Mice	Read	# Mice	Read	# Mice
23SrRNA	15	5/10	8470	10/10	1160	10/10	297	10/10	33	7/10	338	9/9	206	8/10	46	6/10
16SrRNA	31	7/10	5098	10/10	1708	10/10	516	9/10	31	8/10	430	9/9	378	8/10	56	8/10
omcB	0	0/10	483	10/10	148	10/10	16	7/10	0	0/10	29	8/9	15	4/10	2	1/10
hctB	0	0/10	29	9/10	5	4/10	0	0/10	0	0/10	3	2/9	2	1/10	0	0/10
ompA	0	0/10	20	7/10	10	2/10	0	0/10	0	0/10	0	0/9	0	0/10	0	0/10
groEL_1	0	0/10	7	3/10	5	3/10	2	2/10	0	0/10	0	0/9	1	1/10	0	0/10
ftsH	0	0/10	4	3/10	6	4/10	0	0/10	0	0/10	0	0/9	0	0/10	0	0/10
pmpF	0	0/10	5	4/10	1	1/10	0	0/10	0	0/10	0	0/9	0	0/10	0	0/10
CT_858	0	0/10	1	1/10	1	1/10	0	0/10	0	0/10	0	0/9	0	0/10	0	0/10

Average gene expression across time points is shown in Supplementary Figure S2. At -2 dpi,

highly expressed genes were primarily ribosomal proteins (e.g., RPL41, RPS24, RPS27A), reflecting baseline cellular activity. Following infection, a shift in expression patterns was observed, with genes associated with immune and inflammatory responses, such as S100A9 and S100A6, becoming among the most highly expressed at 4 and 6 dpi. Expression levels peaked at 6 dpi, where S100A9 showed the highest expression among all genes. By 11 dpi, several genes that were highly expressed at 6 dpi remained among the top-expressed genes, but their average expression levels were generally lower. The top 10 most highly expressed genes at each time point are summarized in Supplementary Table S1.

Principal component analysis (PCA) of rlog-transformed gene expression data is shown in Figure 4. Samples primarily clustered by time point, with -2 dpi samples grouping closely together, consistent with the correlation heatmaps. Separation between naive and vaccinated samples was also observed along the PC2 axis, with vaccinated samples generally exhibiting higher PC2 values. This suggests that infection time point was the main source of variation, while vaccination status also contributed to gene expression differences.

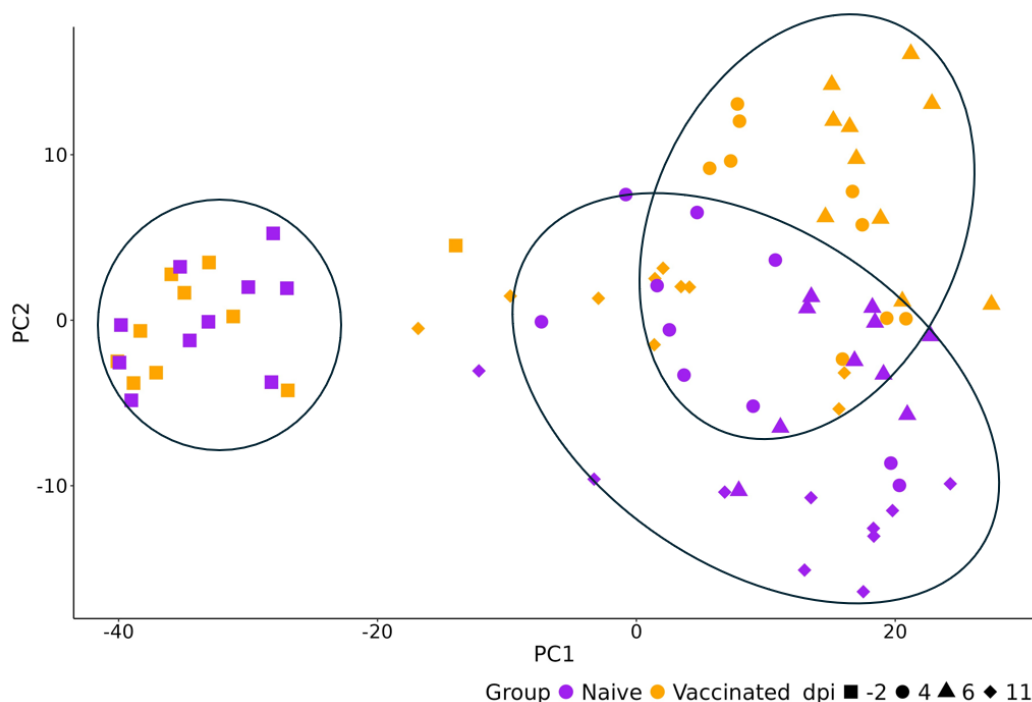


Figure 4: Principal component analysis (PCA) of rlog-transformed gene expression data. Samples clustered primarily by time point, with -2 dpi samples forming a distinct cluster. Separation between naive and vaccinated samples was also observed along PC2. Colors indicate group, and shapes indicate time point.

Statistical testing supported these patterns. MANOVA results (Supplementary Table S2) showed that both group and day had significant effects on overall gene expression variation (Pillai's trace, $p < 0.001$ for both). Univariate ANOVA further indicated that variation along PC1 was primarily driven by time, while PC2 captured variation associated with both group and time, with a stronger contribution from group.

Differential expression

Differential gene expression between vaccinated and naive mice at each time point is shown in Figure 5. The number of significantly differentially expressed genes varied across time points, increasing from 162 genes at 4 dpi to 782 genes at 6 dpi, then declining to 298 genes by 11 dpi, reflecting a pattern consistent with a peak immune response in vaccinated mice at 6 dpi followed by resolution.

At -2 dpi, no genes were significantly differentially expressed between vaccinated and naive mice after multiple testing correction, indicating comparable baseline gene expression profiles prior to infection.

At 4 dpi, the volcano plot based on adjusted p-values, showed clear differential expression between vaccinated and naive mice, with multiple genes upregulated in vaccinated mice. Genes such as *S100a8*, *Il17a*, and *Cd3e* were among those upregulated in vaccinated mice, many of which are associated with immune and inflammatory responses. In contrast, fewer genes were upregulated in naive mice, with generally smaller fold changes.

At 6 dpi, differential expression became more pronounced compared to 4 dpi, with a large increase in the number of significantly regulated genes. Upregulated genes in vaccinated mice included *Pthlh*, *Uox*, *Rnase2b*, *Spink4*, and *Chil4*, which are associated with host defense, epithelial barrier function, and inflammatory responses. In contrast, naive mice also showed upregulation of immune-related genes, including *Gzma*, *Gzmb*, and *Ido1*.

At 11 dpi, differential expression between vaccinated and naive mice persisted. The vaccinated group showed a smaller number of strongly upregulated genes, including *Uox*, *Spink4*, and *Krt17*, suggesting a more resolved or regulated response. In contrast, the naive group exhibited a broader distribution of significantly upregulated genes, including *Ifi7*, *Il2ra*, and *Tnfrsf4*, many of which are associated with immune activation. This pattern indicates a continued and more widespread

inflammatory response in naive mice at this later stage of infection.

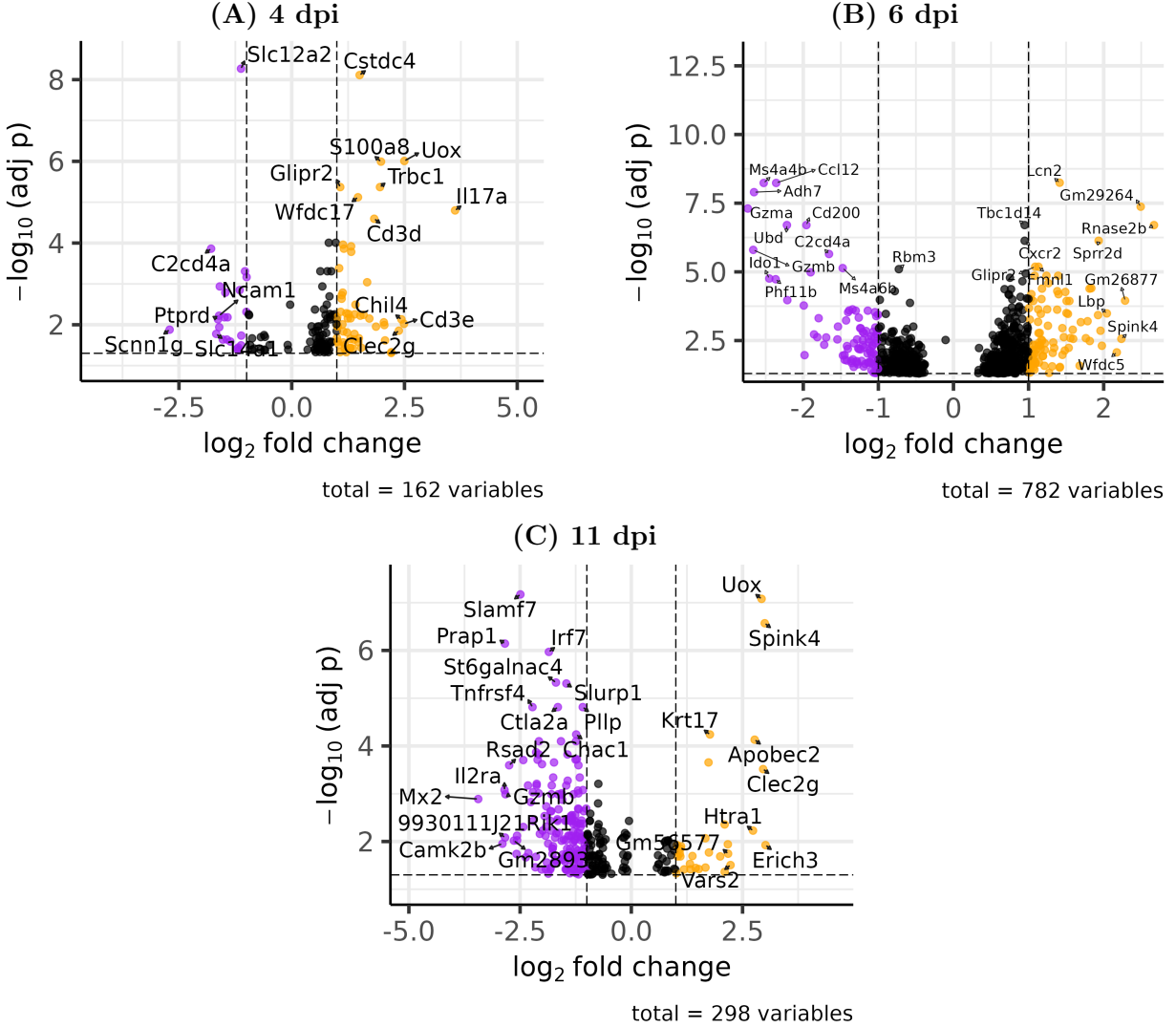


Figure 5: Volcano plots showing differential gene expression between vaccinated and naive mice at 4, 6, and 11 dpi. (A) 4 dpi, (B) 6 dpi, and (C) 11 dpi. The x-axis represents $\log_2$ fold change (vaccinated vs naive), and the y-axis represents $-\log_{10}(\text{adjusted p-value})$. Dashed lines indicate thresholds for statistical significance (adjusted p-value < 0.05) and effect size ($|\log_2 \text{fold change}| > 1$). Selected genes are labeled.

Discussion

This study used longitudinal cervical swab-based dual RNA sequencing to characterize local host and pathogen transcriptional responses to *C. muridarum* infection in vaccinated and naive mice, with the goal of identifying transcriptional signatures associated with vaccine-induced protection.

Vaccinated mice exhibited lower chlamydial gene counts at all post-infection time points, indicat-

ing reduced bacterial burden following vaccination. At baseline (-2 dpi), no significant differences in host gene expression were observed between groups, confirming comparable starting conditions. Following challenge, gene expression patterns diverged between groups. Vaccinated mice showed an early and focused immune response beginning at 4 dpi, peaking at 6 dpi, and declining by 11 dpi. In contrast, naive mice showed fewer and weaker transcriptional differences at 4 dpi, with a more substantial immune response becoming apparent at 6 dpi and persisting through 11 dpi. This pattern is consistent with the established immunological principle that prior antigen exposure generates immunological memory, enabling a more rapid and controlled recall response upon re-exposure to the pathogen, while unvaccinated hosts mount a delayed and more prolonged inflammatory response [Pulendran and Ahmed, 2011].

The transcriptional profiles observed in vaccinated mice provide insight into potential mechanisms of protection. Early upregulation of *S100a8* and *S100a9* suggests activation of innate inflammatory pathways associated with neutrophil recruitment and antimicrobial responses at mucosal surfaces [Vogl et al., 2014]. Concurrent upregulation of *Il17a* indicates involvement of IL-17-mediated mucosal immunity, which plays a key role in recruiting neutrophils and enhancing epithelial defense against bacterial infection [Kolls and Khader, 2010]. The presence of *Cd3e* further supports engagement of T cell-mediated responses [Smith-Garvin et al., 2009]. Together, these findings suggest that CPAF-based vaccination promotes a coordinated immune response involving early innate activation and rapid recruitment of adaptive immunity at the site of infection.

These observations are consistent with prior studies demonstrating that protection against Chlamydia infection is primarily mediated by CD4 T cell responses. In particular, CPAF + ADU-S100 vaccination has been shown to induce effector CD4 T cells producing cytokines such as IL-17A and IFN γ , leading to reduced bacterial burden following challenge [Poston et al., 2025]. Our findings extend this work by demonstrating that these protective immune responses are reflected in local transcriptional changes detectable through cervical swab sampling. The detection of IL-17-associated transcriptional activity in vaccinated mice supports the role of mucosal Th17 responses in mediating protection.

In contrast, naive mice exhibited a broader and more sustained transcriptional response at later time points, likely reflecting ongoing bacterial replication and continued inflammatory stimulation. This suggests that stronger late-stage immune activation does not necessarily correspond to effective

protection, but instead may indicate delayed control of infection and prolonged tissue exposure to inflammatory signaling. This distinction is important because *Chlamydia* infections can be asymptomatic yet still lead to tissue damage and reproductive tract pathology. Therefore, persistent inflammatory transcriptional activity in unprotected hosts may reflect ongoing immunopathology rather than an effective protective response.

Importantly, this study demonstrates the feasibility of applying probe-based dual RNA sequencing to longitudinal cervical swab samples for the analysis of host–pathogen interactions. Because swab collection is minimally invasive and can be repeated over time, this strategy enables dynamic characterization of mucosal immune responses without the need for terminal tissue collection. It may therefore provide a practical platform for studying infection dynamics and vaccine responses in mucosal disease models.

Several limitations should be considered. First, low levels of chlamydial transcripts were detected at -2 dpi despite the absence of infection, potentially reflecting background signal or probe cross-reactivity with conserved bacterial rRNA sequences. Second, the sample size of 10 mice per group is modest, and larger cohorts will be necessary to validate the identified transcriptional signatures. Third, the use of a single inbred mouse strain may limit generalizability across genetic backgrounds and to human infection. Fourth, differential expression was analyzed separately at each time point, with multiple-testing correction applied within rather than across time points, and without directly modeling group-by-time interactions.

Beyond these limitations, future studies should extend these findings by using mixed-effects models with group-by-time interaction terms to directly assess temporal transcriptional dynamics between groups. Pathway-level analyses, such as gene set enrichment analysis or Reactome-based approaches, could further identify coordinated biological processes underlying vaccine-induced protection. In addition, the use of genetically diverse mouse populations, such as Collaborative Cross (CC) mice, would help evaluate the robustness of transcriptional signatures across host genetic variation. The swab-based dual RNA-seq framework described here may also be applied to other sexually transmitted infection models, enabling minimally invasive investigation of local immune responses. Finally, determining whether these transcriptional signatures are conserved in human infection will be critical for translating these findings into clinically relevant biomarkers.

Overall, this study establishes a longitudinal swab-based transcriptional framework for evaluat-

ing mucosal vaccine responses and identifies candidate host and pathogen markers that may inform the design of future chlamydia vaccine studies.

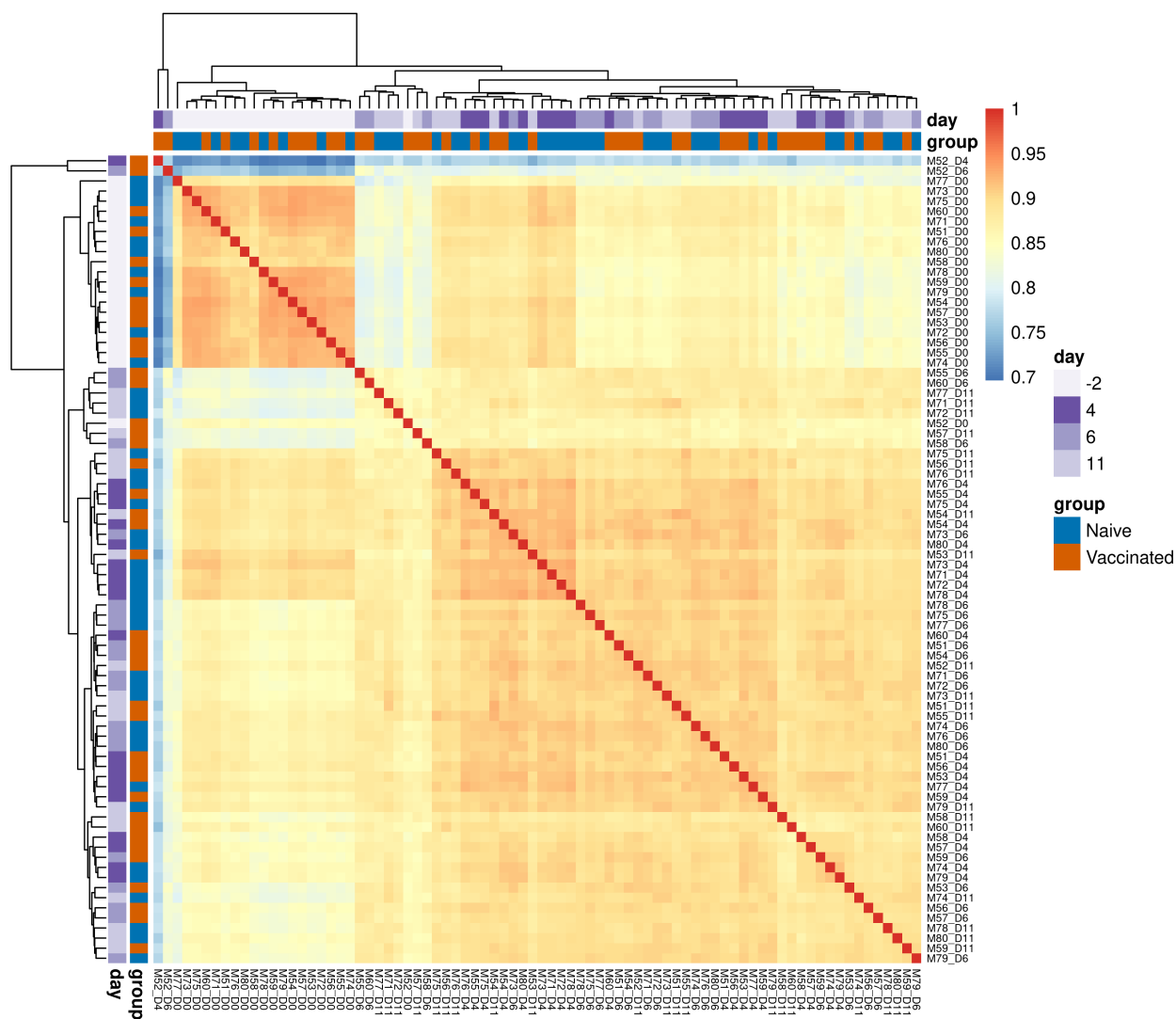
References

- [Benjamini and Hochberg, 1995] Benjamini, Y. and Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57:289–300.
- [Broder et al., 2024] Broder, K. C., Matrosova, V. Y., Tkavc, R., Gaidamakova, E. K., Ho, L. T. V. T., Macintyre, A. N., Soc, A., Diallo, A., Darnell, S. C., Bash, S., Daly, M. J., Jerse, A. E., and Liechti, G. W. (2024). Irradiated whole cell chlamydia vaccine confers significant protection in a murine genital tract challenge model. *npj Vaccines*, 9.
- [CDC, 2024] CDC (2024). Sexually transmitted infections surveillance, 2023.
- [CDC, 2025] CDC (2025). About chlamydia.
- [Gogoi et al., 2020] Gogoi, H., Mansouri, S., and Jin, L. (2020). The age of cyclic dinucleotide vaccine adjuvants. *Vaccines*, 8(3):453.
- [Kari et al., 2009] Kari, L., Whitmire, W. M., Crane, D. D., Reveneau, N., Carlson, J. H., Goheen, M. M., Peterson, E. M., Pal, S., de la Maza, L. M., and Caldwell, H. D. (2009). Chlamydia trachomatis native major outer membrane protein induces partial protection in non-human primates: Implication for a trachoma transmission blocking vaccine. *Journal of Immunology*, 182:8063–8070.
- [Karunakaran et al., 2010] Karunakaran, K. P., Yu, H., Foster, L. J., and Brunham, R. C. (2010). Development of a chlamydia trachomatis t cell vaccine. *Human Vaccines*, 6:676–680.
- [Kolls and Khader, 2010] Kolls, J. K. and Khader, S. A. (2010). The role of th17 cytokines in primary mucosal immunity. *Cytokine & Growth Factor Reviews*, 21:443–448.
- [Kumar et al., 2021] Kumar, S., Chesson, H., and Gift, T. L. (2021). Estimating the direct medical costs and productivity loss of outpatient chlamydia and gonorrhea treatment. *Sexually Transmitted Diseases*, 48(2):e18–e21.
- [Love et al., 2014] Love, M. I., Huber, W., and Anders, S. (2014). Moderated estimation of fold change and dispersion for rna-seq data with deseq2. *Genome Biology*, 15:550.

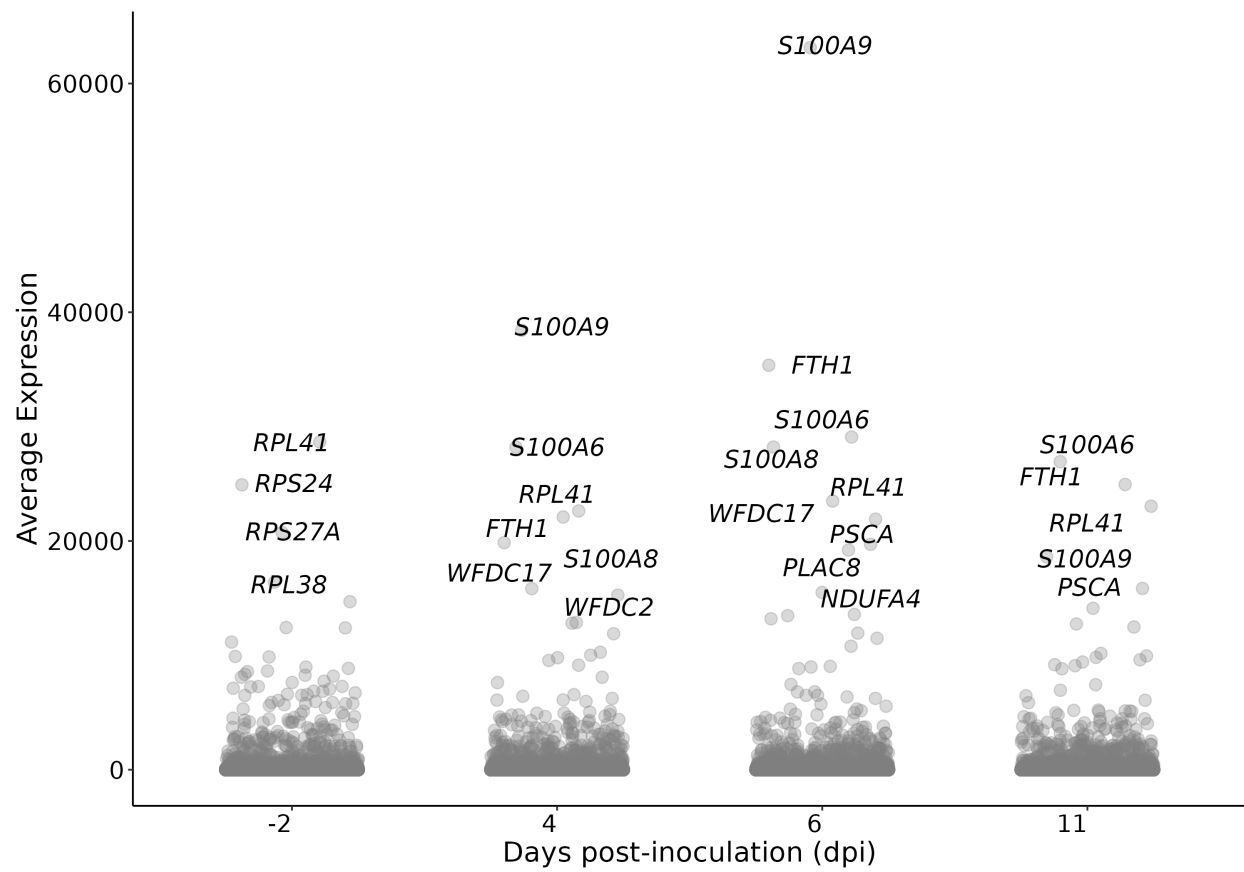
- [Mabonga et al., 2021] Mabonga, E., Manabe, Y. C., Elbireer, A., Mbazira, J. K., Nabaggala, M. S., Kiragga, A., Kisakye, J., Gaydos, C. A., Taylor, C., and Parkes-Ratanshi, R. (2021). Prevalence and predictors of asymptomatic *Chlamydia trachomatis* and *Neisseria gonorrhoeae* in a ugandan population most at risk of hiv transmission. *International Journal of STD & AIDS*, 32:510–516.
- [Meyer, 2016] Meyer, T. (2016). Diagnostic procedures to detect chlamydia trachomatis infections. *Microorganisms*, 4:25.
- [Morrison and Caldwell, 2002] Morrison, R. P. and Caldwell, H. D. (2002). Immunity to murine chlamydial genital infection. *Infection and Immunity*, 70:2741–2751.
- [Murthy et al., 2009] Murthy, A. K., Guentzel, M. N., Zhong, G., and Arulanandam, B. P. (2009). Chlamydial protease-like activity factor—insights into immunity and vaccine development. *Journal of Reproductive Immunology*, 83:179–184.
- [Poston et al., 2025] Poston, T. B., Girardi, J., Kim, M., Zwarycz, P., Polson, A. G., Yount, K. S., Hanlan, C., Jaras Salas, I., Lammert, S. M., Arroyo, D., Bruno, T., Wu, M., Rozzelle, J., Fairman, J., Esser-Kahn, A. P., and Darville, T. (2025). Intranasal immunization with cpaf combined with adu-s100 induces an effector cd4 t cell response and reduces bacterial burden following intravaginal infection with *Chlamydia muridarum*. *Vaccine*, 43(Pt 1):126526.
- [Proctor et al., 2024] Proctor, J., Stadler, M., Cortes, L. M., Brodsky, D., Poisson, L., Gerdts, V., Smirnov, A. I., Smirnova, T. I., Barua, S., Leahy, D., Beagley, K. W., Harris, J. M., Darville, T., and Käser, T. (2024). A triadj-adjuvanted chlamydia trachomatis cpaf protein vaccine is highly immunogenic in pigs. *Vaccines*, 12(4):423.
- [Pulendran and Ahmed, 2011] Pulendran, B. and Ahmed, R. (2011). Immunological mechanisms of vaccination. *Nature Immunology*, 12:509–517.
- [Robinson et al., 2010] Robinson, M. D., McCarthy, D. J., and Smyth, G. K. (2010). edgeR: A Bioconductor package for differential expression analysis of digital gene expression data. *Bioinformatics*, 26(1):139–140.

- [Schautteet et al., 2011] Schautteet, K., De Clercq, E., and Vanrompay, D. (2011). Chlamydia trachomatis vaccine research through the years. *Infectious Diseases in Obstetrics and Gynecology*, 2011:963513.
- [Smith-Garvin et al., 2009] Smith-Garvin, J. E., Koretzky, G. A., and Jordan, M. S. (2009). T cell activation. *Annual Review of Immunology*, 27:591–619.
- [Stary, 2015] Stary, G. e. a. (2015). A mucosal vaccine against chlamydia trachomatis generates two waves of protective memory t cells. *Science*, 348:aaa8205.
- [Vogl et al., 2014] Vogl, T., Eisenblätter, M., Völler, T., Zenker, S., Hermann, S., van Lent, P., Faust, A., Geyer, C., Petersen, B., Roebrock, K., Schäfers, M., Bremer, C., and Roth, J. (2014). Alarmin s100a8/s100a9 as a biomarker for molecular imaging of local inflammatory activity. *Nature Communications*, 5.
- [WHO, 2024] WHO (2024). Chlamydia.
- [Zhu et al., 2018] Zhu, A., Ibrahim, J. G., and Love, M. I. (2018). Heavy-tailed prior distributions for sequence count data: removing the noise and preserving large differences. *Bioinformatics*, 35:2084–2092.

Supplementary Figures



Supplementary Figure S1: Sample-to-sample Spearman correlation heatmap based on $\log_2$ CPM-normalized gene expression data, showing clustering patterns consistent with those observed in the Pearson correlation heatmap.



Supplementary Figure S2: Distribution of average gene expression across time points. Each point represents a gene, with highly expressed genes labeled.

Supplementary Tables

Supplementary Table S1: Top 10 most highly expressed genes at each time point based on average expression across samples.

Day (dpi)	Gene	Average Expression
-2	Rpl41	28664.35
-2	Rps24	24913.95
-2	Rps27a	20746.20
-2	Rpl38	16399.30
-2	Rpl37	14690.40
-2	S100a6	12429.90
-2	Sec61g	12403.85
-2	Rps20	11163.20
-2	Rpl39	9905.85
-2	Wfdc2	9860.75
4	S100a9	38424.25
4	S100a6	28209.15
4	Rpl41	22627.35
4	Fth1	22098.55
4	S100a8	19857.00
4	Wfdc17	15839.35
4	Wfdc2	15263.80
4	Ndufa4	12870.90
4	Rps27a	12815.35
4	Rps24	11893.20
6	S100a9	63140.10
6	Fth1	35366.75
6	S100a6	29091.05
6	S100a8	28215.45
6	Rpl41	23494.10
6	Wfdc17	21907.35
6	Psca	19710.45
6	Plac8	19209.80
6	Ndufa4	15500.35
6	Rps24	13582.80
11	S100a6	26939.30
11	Fth1	24939.05
11	Rpl41	23040.20
11	S100a9	18619.75
11	Psca	15851.65
11	Ndufa4	14113.25
11	Rps24	12747.05
11	Rps27a	12479.00
11	Rpl38	10168.30
11	Anxa1	9952.10

Supplementary Table S2: MANOVA and follow-up ANOVA results for principal components derived from rlog-transformed gene expression data.

Analysis	Effect	Statistic	F-value	p-value
MANOVA (PC1 and PC2 jointly)	group	Pillai = 0.399	24.21	8.60×10^{-9}
MANOVA (PC1 and PC2 jointly)	day	Pillai = 1.209	37.73	$< 2.2 \times 10^{-16}$
ANOVA for PC1	group	—	0.042	0.839
ANOVA for PC1	day	—	154.58	$< 2.2 \times 10^{-16}$
ANOVA for PC2	group	—	45.14	3.27×10^{-9}
ANOVA for PC2	day	—	13.17	5.60×10^{-7}

Supplementary Table S3: Top differentially expressed genes at day 4 post-infection between vaccinated and naive mice.

Upregulated in Naive			Upregulated in Vaccinated		
Gene	log2FC	padj	Gene	log2FC	padj
Scnn1g	-2.71	0.013	Il17a	3.62	< 0.001
C2cd4a	-1.79	< 0.001	Uox	2.50	< 0.001
Slc14a1	-1.67	0.017	Cd3e	2.49	0.009
Ptprd	-1.63	0.007	Chil4	2.43	0.007
Ncam1	-1.61	0.006	Clec2g	2.38	0.014
S100g	-1.61	0.012	Cd6	2.21	0.049
Ank	-1.60	0.001	Trac	2.07	0.024
Ly6g6c	-1.53	0.023	Cxcr6	2.05	0.010
Pdzrn4	-1.49	0.007	S100a8	1.98	< 0.001
Adh7	-1.47	0.002	Trbc1	1.96	< 0.001

Supplementary Table S4: Top differentially expressed genes at day 6 post-infection between vaccinated and naive mice.

Upregulated in Naive			Upregulated in Vaccinated		
Gene	log2FC	padj	Gene	log2FC	padj
Gzma	-2.74	< 0.001	Pthlh	2.89	< 0.001
Gzmb	-2.67	< 0.001	Uox	2.86	< 0.001
Adh7	-2.66	< 0.001	Rnase2b	2.67	< 0.001
Ms4a4b	-2.53	< 0.001	Gm29264	2.49	< 0.001
Ido1	-2.45	< 0.001	Gm26877	2.29	< 0.001
Phf11b	-2.37	< 0.001	Spink4	2.24	0.003
Ccl12	-2.36	< 0.001	Wfdc5	2.17	0.009
Ubd	-2.22	< 0.001	Lbp	2.04	< 0.001
S100g	-2.21	< 0.001	Chil4	1.98	0.005
Akr1c18	-1.99	< 0.001	Rnase1	1.96	0.001

Supplementary Table S5: Top differentially expressed genes at day 11 post-infection between vaccinated and naive mice.

Upregulated in Naive			Upregulated in Vaccinated		
Gene	log2FC	padj	Gene	log2FC	padj
Mx2	-3.44	0.001	Erich3	3.02	0.012
Camk2b	-2.90	0.011	Spink4	3.00	< 0.001
Il2ra	-2.85	< 0.001	Clec2g	2.96	< 0.001
Prap1	-2.84	< 0.001	Uox	2.93	< 0.001
9930111J21Rik1	-2.84	0.008	Apobec2	2.77	< 0.001
Gzmb	-2.83	0.001	Htra1	2.74	0.006
Rsad2	-2.75	< 0.001	Vars2	2.23	0.031
Gm28935	-2.64	0.010	Gm56577	2.17	0.018
Gimap1os	-2.58	0.018	Gpc3	2.17	0.011
Pdcd1	-2.57	0.008	Hmga2	2.10	0.043